

PROFESSIONAL SUMMARY

System Engineer with 3.5+ years of experience building ML-based anomaly detection pipelines, LLM-integrated automation, and RAG-based Q&A systems. Skilled in Python, Azure, and the Microsoft Power Platform, with hands-on experience taking GenAI prototypes from data ingestion through retrieval and deployment. Actively expanding into FastAPI-based model serving and AWS to broaden delivery beyond an Azure-centric stack.

EXPERIENCE

Tata Consultancy Services

July 2022 – Present

System Engineer

- Built ML-based anomaly detection pipelines in Python to analyze sensor data, improving anomaly identification accuracy by **~18%**.
- Developed a Streamlit dashboard to visualize trends, anomaly alerts, and risk indicators, cutting manual inspection time by **~40%**.
- Integrated LLMs to auto-generate insights and summaries for anomaly events, reducing manual analysis workload by **50–60%**.
- Developed a Power Apps application to centralize idea/innovation tracking, incorporating structured approval and validation workflows.
- Built an AI-driven chatbot using Microsoft Copilot, Power Automate, and Dataverse to automate ServiceNow ticket creation and answer FAQs, reducing manual workload by **30–40%**.
- Implemented an automated NPS feedback workflow in Microsoft Teams, increasing response rates from **3% to 65%**.
- Supported Oracle Database activities including performance tuning, backup/recovery, and query optimization.

PROJECTS

Document Q&A Chatbot | RAG

- Developed a RAG-based Q&A system using LangChain, LanceDB for vector storage, and Cohere embeddings for semantic search.
- Implemented document ingestion, text extraction, chunking, and embedding generation for PDFs/DOCX.
- Designed a semantic retrieval pipeline enabling high-accuracy context matching through vector similarity search.
- Integrated Groq-hosted LLMs for grounded answer generation.
- Built a lightweight interactive UI for querying documents and displaying sourced outputs.

Multiplayer Chess Platform

- Building a chess platform from scratch in Node.js using WebSocket (ws) for real-time multiplayer gameplay and chess.js for move validation.
- Designing the architecture to support future bots that mimic grandmaster playing styles via CNNs trained on PGN game data.

TECHNICAL SKILLS

Programming Languages : Python, C++, SQL

Machine Learning & AI : Pandas, NumPy, Machine Learning Algorithms, Model Evaluation & Deployment

GenAI & RAG : LangChain, RAG Pipelines, Vector Search (LanceDB), Embeddings (Cohere), LLM Integration (Groq)

Cloud & DevOps : Microsoft Azure, Git, CI/CD Pipelines

Microsoft Power Platform : Power Automate, Power Apps

Software Development : OOP, DSA, REST APIs

EDUCATION

B.Tech – Computer Science and Engineering

June 2018 - July 2022

Camellia School of Engineering and Technology

Intermediate (10+2)

2018

Kendriya Vidyalaya No. 4, Bathinda Cantt.

CERTIFICATIONS

- AZ-900: Azure Fundamentals
- AI-103: Develop AI Apps and Agents on Azure (Pursuing)

ACHIEVEMENTS

- NPTEL Course Topper: Problem Solving Through Programming in C (2019)